Sri Lanka Institute of Information Technology
Year 3 – Semester 1
Network Design and Management

Practical 06 – LDAP

Introduction to OpenLDAP

OpenLDAP is a free, open-source implementation of the Lightweight Directory Access Protocol (LDAP) developed by the OpenLDAP project. It is a platform-independent protocol that provides a hierarchical directory service for managing users, groups, and network resources across Linux/Unix, Windows, and other systems.

Prerequisites:

1. Make sure both server Centos and client are accessible.
2. Make an entry of each host in /etc/hosts for name resolution or Configure it in DNS to resolve the IP, if you use server name instead of IP address.

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1. Network Preparation & Service Management

Before beginning the LDAP installation, ensure your network environment is clean and correctly configured for external package downloads.

1. Stop all the running services.

[root@mlb-dc1-centos9 ~]# service dhcpd stop
[root@mlb-dc1-centos9 ~]# service named stop

```
[root@mlb-dc1-centos9 ~]# service dhcpd stop
Redirecting to /bin/systemctl stop dhcpd.service
[root@mlb-dc1-centos9 ~]# service named stop
Redirecting to /bin/systemctl stop named.service
```

Note: Root privilege is needed for the terminal.

2. Set VMware network settings to vmnet8-NAT and make your server IP settings to obtain IP addresses automatically.

3. Restart the network service

Note: You should have a look at your IP address to check whether the IP is within the network range defined by vmnet8-NAT configured DHCP server.

2. Installing LDAP Packages and MigrationTools

In this section, we enable the EPEL repository to install the OpenLDAP server and client tools, then manually set up the migration scripts required for user management.

1. Install OpenLDAP Server and Clients

Use the dnf package manager to install the core LDAP components.

Activate → NAT & Deactivate → VMnet2

[root@mlb-dc1-centos9 ~]# dnf install -y epel-release

[root@mlb-dc1-centos9 ~]# dnf install -y openldap openldap-servers openldap-clients

```
[root@mlb-dc1-centos9 ~]# dnf install -y epel-release
Last metadata expiration check: 0:29:57 ago on Sat 21 Mar 2026 12:34:41 AM EDT.
Dependencies resolved.
================================================================================
 Package                 Architecture Version         Repository           Size
================================================================================
Installing:
 epel-release            noarch       9-7.el9         extras-common        19 k
Installing weak dependencies:
 epel-next-release       noarch       9-7.el9         extras-common       8.1 k

Transaction Summary
================================================================================
Install  2 Packages
```

```
Upgraded:
  openldap-2.6.8-4.el9.x86_64
Installed:
  openldap-clients-2.6.8-4.el9.x86_64    openldap-servers-2.6.8-2.el9.x86_64

Complete!
[root@mlb-dc1-centos9 ~]# dnf install -y openldap openldap-servers openldap-clien
ts
Last metadata expiration check: 0:01:22 ago on Sat 21 Mar 2026 01:07:43 AM EDT.
Package openldap-2.6.8-4.el9.x86_64 is already installed.
Package openldap-servers-2.6.8-2.el9.x86_64 is already installed.
Package openldap-clients-2.6.8-4.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@mlb-dc1-centos9 ~]#
```

2. Setup MigrationTools for CentOS 9

Since the migrationtools package is not available as a standard RPM in CentOS 9, we must manually download the Perl scripts.

1. Install Prerequisites

Ensure that wget (for downloading) and perl (to run the scripts) are installed on your system.

[root@mlb-dc1-centos9 ~]# dnf install wget perl -y

```
[root@mlb-dc1-centos9 ~]# dnf install wget perl -y
Last metadata expiration check: 2:17:51 ago on Sat 21 Mar 2026 01:07:43 AM EDT.
Package wget-1.21.1-8.el9.x86_64 is already installed.
Dependencies resolved.
==========================================================================================================
 Package                                Architecture     Version                        Repository          Size
==========================================================================================================
Installing:
 perl                                   x86_64           4:5.32.1-483.el9               appstream          8.2 k
Upgrading:
 glibc                                  x86_64           2.34-256.el9                   baseos             2.0 M
 glibc-all-langpacks                    x86_64           2.34-256.el9                   baseos              18 M
 glibc-common                           x86_64           2.34-256.el9                   baseos             307 k
 glibc-gconv-extra                      x86_64           2.34-256.el9                   baseos             1.7 M
 glibc-langpack-en                      x86_64           2.34-256.el9                   baseos             661 k
 perl-Errno                             x86_64           1.30-483.el9                   appstream           15 k
 perl-interpreter                       x86_64           4:5.32.1-483.el9               appstream           70 k
 perl-libs                              x86_64           4:5.32.1-483.el9               appstream          2.2 M
Installing dependencies:
```

2. Create Directory and Download Scripts

Create a dedicated directory for the tools and download the official migration script archive.

[root@mlb-dc1-centos9 ~]# mkdir -p /usr/share/migrationtools
[root@mlb-dc1-centos9 ~]# cd /usr/share/migrationtools

```
[root@mlb-dc1-centos9 ~]# mkdir -p /usr/share/migrationtools
[root@mlb-dc1-centos9 ~]# cd /usr/share/migrationtools
```

3. Download the migration tools archive

[root@mlb-dc1-centos9 migrationtools]# wget http://archive.ubuntu.com/ubuntu/pool/universe/m/migrationtools/migrationtools_47.orig.tar.gz

```
[root@mlb-dc1-centos9 migrationtools]# wget http://archive.ubuntu.com/ubuntu/pool/universe/m/migrationtools/migrati
ontools_47.orig.tar.gz
--2026-03-21 03:28:46--  http://archive.ubuntu.com/ubuntu/pool/universe/m/migrationtools/migrationtools_47.orig.tar
.gz
Resolving archive.ubuntu.com (archive.ubuntu.com)... 91.189.92.22, 91.189.92.24, 185.125.190.81, ...
Connecting to archive.ubuntu.com (archive.ubuntu.com)|91.189.92.22|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 21284 (21K) [application/x-gzip]
Saving to: 'migrationtools_47.orig.tar.gz'

migrationtools_47.orig.tar.g 100%[=============================================>]  20.79K  30.1KB/s    in 0.7s

2026-03-21 03:28:48 (30.1 KB/s) - 'migrationtools_47.orig.tar.gz' saved [21284/21284]
```

4. Extract the Files

Extract the archive and use --strip-components=1 to place the scripts directly into your current directory.

[root@mlb-dc1-centos9 migrationtools]# tar -zxvf migrationtools_47.orig.tar.gz --strip-components=1

```
[root@mlb-dc1-centos9 migrationtools]# tar -zxvf migrationtools_47.orig.tar.gz --strip-components=1
```

3. Generating the LDAP Administrator Password.

[root@mlb-dc1-centos9 ~]# slappasswd
New password:
Re-enter new password:
{SSHA}bHSiwuPJEypHS6zHSE2Uy7M69sQjmkPL

```
[root@mlb-dc1-centos9 ~]# slappasswd
New password:
Re-enter new password:
{SSHA}wXXIqvU1L9IDhx7K2eK5MepmIRmHigzu
```

Copy the entire output string, including the {SSHA} prefix. You will need this string in the next step to define the olcRootPW in your server configuration.

4. Configuring the OpenLDAP Server Database

The OpenLDAP server in CentOS 9 uses a dynamic configuration system (OLC) located in /etc/openldap/slapd.d/. We must modify the database configuration file to define our domain and administrative credentials.

1. Navigate to the Configuration Directory

   The database-specific settings are stored in the cn=config directory.

   [root@mlb-dc1-centos9 ~]# cd /etc/openldap/slapd.d/cn=config

```
[root@mlb-dc1-centos9 ~]# cd /etc/openldap/slapd.d/cn=config
[root@mlb-dc1-centos9 cn=config]#
```

2. Edit the Database Configuration File

   Open the mdb (Memory-Mapped Database) configuration file.

   Note: In CentOS 9, the file is typically named olcDatabase={2}mdb.ldif.

   [root@mlb-dc1-centos9 cn=config]# vi olcDatabase={2}mdb.ldif

```
[root@mlb-dc1-centos9 cn=config]# vi olcDatabase={2}mdb.ldif
[root@mlb-dc1-centos9 cn=config]#
```

3. Update Domain and Admin Variables

   Locate and change the following lines to match your domain (csa.lk)

   olcSuffix: dc=csa,dc=lk

   olcRootDN: cn=Manager,dc=csa,dc=lk

```
# AUTO-GENERATED FILE - DO NOT EDIT!! Use ldapmodify.
# CRC32 132378b8
dn: olcDatabase={2}mdb
objectClass: olcDatabaseConfig
objectClass: olcMdbConfig
olcDatabase: {2}mdb
olcDbDirectory: /var/lib/ldap
olcSuffix: dc=csa,dc=lk
olcRootDN: cn=Manager,dc=csa,dc=lk
olcDbIndex: objectClass eq,pres
olcDbIndex: ou,cn,mail,surname,givenname eq,pres,sub
structuralObjectClass: olcMdbConfig
entryUUID: ca48d962-b92f-1040-95c5-b5d284c6b3dc
creatorsName: cn=config
createTimestamp: 20260321050849Z
entryCSN: 20260321050849.285531Z#000000#000#000000
modifiersName: cn=config
modifyTimestamp: 20260321050849Z
```

4. Add Administrative Password and TLS Paths

   Add the following lines to the end of the file. Replace the olcRootPW with the hashed

password you generated in Part 3.
olcRootPW: {SSHA}bHSiwuPJEypHS6zHSE2Uy7M69sQjmkPL
olcTLSCertificateFile: /etc/pki/tls/certs/csaldap.pem
olcTLSCertificateKeyFile: /etc/pki/tls/certs/csaldapkey.pem

```
# AUTO-GENERATED FILE - DO NOT EDIT!! Use ldapmodify.
# CRC32 132378b8
dn: olcDatabase={2}mdb
objectClass: olcDatabaseConfig
objectClass: olcMdbConfig
olcDatabase: {2}mdb
olcDbDirectory: /var/lib/ldap
olcSuffix: dc=csa,dc=lk
olcRootDN: cn=Manager,dc=csa,dc=lk
olcDbIndex: objectClass eq,pres
olcDbIndex: ou,cn,mail,surname,givenname eq,pres,sub
structuralObjectClass: olcMdbConfig
entryUUID: ca48d962-b92f-1040-95c5-b5d284c6b3dc
creatorsName: cn=config
createTimestamp: 20260321050849Z
entryCSN: 20260321050849.285531Z#000000#000#000000
modifiersName: cn=config
modifyTimestamp: 20260321050849Z
olcRootPW: {SSHA}wXXIqvU1L9IDhx7K2eK5MepmIRmHigzu
olcTLSCertificateFile: /etc/pki/tls/certs/csaldap.pem
olcTLSCertificateKeyFile: /etc/pki/tls/certs/csaldapkey.pem
```

Note: The olcSuffix defines the domain name for the LDAP directory. The olcRootDN acts as the "root" user for the LDAP server, granting unrestricted administrative access to the directory.

5. Providing Monitor Privileges and Verification

The monitor database allows for tracking the status and performance of the LDAP server. You must restrict access to this information to the administrative user.

1. Configure Monitor Access

   Open the monitor database configuration file and update the access controls to match your domain.
   [root@mlb-dc1-centos9 cn=config]# vi olcDatabase={1}monitor.ldif

```
[root@mlb-dc1-centos9 cn=config]# vi olcDatabase={1}monitor.ldif
```

   Locate the line starting with olcAccess and update it as follows:
   olcAccess: {0}to * by dn.base="gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth" read by dn.base="cn=Manager,dc=csa,dc=lk" read by * none

```
# AUTO-GENERATED FILE - DO NOT EDIT!! Use ldapmodify.
# CRC32 0b007869
dn: olcDatabase={1}monitor
objectClass: olcDatabaseConfig
olcDatabase: {1}monitor
olcAccess: {0}to * by dn.base="gidNumber=0+uidNumber=0,cn=peercred,cn=external
 ,cn=auth" read by dn.base="cn=Manager,dc=csa,dc=lk" read by * none
structuralObjectClass: olcDatabaseConfig
entryUUID: fed21392-ba0b-1040-8149-47fe9029a1b3
creatorsName: cn=config
createTimestamp: 20260322072506Z
entryCSN: 20260322072506.706900Z#000000#000#000000
modifiersName: cn=config
modifyTimestamp: 20260322072506Z
```

Note: If no olcAccess directives are specified, the default policy allows all users (authenticated and anonymous) read access. By setting this, we ensure only the system root and the LDAP Manager can read the monitor data.

2. Verify the Configuration

Before starting the service, run a test to check for syntax errors in your LDIF files.

[root@mlb-dc1-centos9 cn=config]# slaptest -u

Expected Output: You will likely see checksum errors such as:

ldif_read_file: checksum error on ".../olcDatabase={1}monitor.ldif"
ldif_read_file: checksum error on ".../olcDatabase={2}mdb.ldif"

config file testing succeeded

```
[root@mlb-dc1-centos9 cn=config]# slaptest -u
ldif_read_file: checksum error on "/etc/openldap/slapd.d/cn=config/olcDatabase={1}monitor.ldif"
ldif_read_file: checksum error on "/etc/openldap/slapd.d/cn=config/olcDatabase={2}mdb.ldif"
config file testing succeeded
```

Important: You can safely ignore the checksum errors. These occur because we manually edited the .ldif files with vi, which changes the file content without updating the CRC32 checksum at the top of the file. As long as the command ends with config file testing succeeded, your configuration is valid.

6. Creating Self-Signed Certificates

In Part 4, we specified certificate paths in our configuration. Now, we must generate those files using OpenSSL to allow for encrypted LDAP (LDAPS) communication.

1. Generate the Certificate and Key

Use the openssl command to create a new 365-day self-signed certificate and a private key.

[root@mlb-dc1-centos9 cn=config]# openssl req -new -x509 -nodes -out /etc/pki/tls/certs/csaldap.pem -keyout /etc/pki/tls/certs/csaldapkey.pem -days 365

```
[root@mlb-dc1-centos9 cn=config]# openssl req -new -x509 -nodes -out /etc/pki/tls/certs/csaldap.pem
-keyout /etc/pki/tls/certs/csaldapkey.pem -days 365
```

2. Provide Company Details

Enter the following information when prompted to identify your server:

Country Name (2 letter code) [XX] : SL
State or Province Name (full name) [ ] : Western
Locality Name (eg, city) [Default City] : Colombo
Organization Name (eg, company) [Default Company Ltd] : CSA
Organizational Unit Name (eg, section) [ ] : CSA-Server
Common Name (eg, your name or your server's hostname) [ ] : mlb-dc1-centos9.csa.lk
Email Address [ ] : root@mlb-dc1-centos9.csa.lk

```
Country Name (2 letter code) [XX]:SL
State or Province Name (full name) []:Western
Locality Name (eg, city) [Default City]:Colombo
Organization Name (eg, company) [Default Company Ltd]:CSA
Organizational Unit Name (eg, section) []:CSA-Server
Common Name (eg, your name or your server's hostname) []:mlb-dc1-centos9.csa.lk
Email Address []:root@mlb-dc1-centos9.csa.lk
```

3. Verify the Certificates

   Check the directory to ensure both files exist with the correct permissions.

   [root@mlb-dc1-centos9 cn=config]# ls -l /etc/pki/tls/certs/csaldap*

   Expected Output

   -rw-r--r--. 1 ldap ldap 1497 Mar 21 11:00 /etc/pki/tls/certs/csaldap.pem
   -rw-------. 1 ldap ldap 1704 Mar 21 11:00 /etc/pki/tls/certs/csaldapkey.pem

```
[root@mlb-dc1-centos9 cn=config]# ls -l /etc/pki/tls/certs/csaldap*
-rw-------. 1 root root 1704 Mar 21 01:48 /etc/pki/tls/certs/csaldapkey.pem
-rw-r--r--. 1 root root 1497 Mar 21 01:49 /etc/pki/tls/certs/csaldap.pem
```

7. Finalizing Permissions

Before starting the OpenLDAP service, we must ensure the ldap system user has full control over the configuration, the database, and the security certificates.

1. Secure the Certificates:

   [root@mlb-dc1-centos9 cn=config]# chown ldap:ldap /etc/pki/tls/certs/csaldap*
   [root@mlb-dc1-centos9 cn=config]# chmod 600 /etc/pki/tls/certs/csaldapkey.pem
   [root@mlb-dc1-centos9 cn=config]# chmod 644 /etc/pki/tls/certs/csaldap.pem

```
[root@mlb-dc1-centos9 cn=config]# chown ldap:ldap /etc/pki/tls/certs/csaldap*
[root@mlb-dc1-centos9 cn=config]# chmod 600 /etc/pki/tls/certs/csaldapkey.pem
[root@mlb-dc1-centos9 cn=config]# chmod 644 /etc/pki/tls/certs/csaldap.pem
```

2. Set Database and Config Ownership:

   [root@mlb-dc1-centos9 cn=config]# chown -R ldap:ldap /var/lib/ldap/
   [root@mlb-dc1-centos9 cn=config]# chown -R ldap:ldap /etc/openldap/slapd.d/

```
[root@mlb-dc1-centos9 cn=config]# chown -R ldap:ldap /var/lib/ldap/
[root@mlb-dc1-centos9 cn=config]# chown -R ldap:ldap /etc/openldap/slapd.d/
```

8. Enabling and Starting the OpenLDAP Service

With the configuration, certificates, and permissions in place, we can now start the LDAP daemon and verify that it is listening for connections.

1. Enable and Start the SLAPD Service

Use systemctl to start the service and ensure it launches automatically at system boot.

Activate → VMnet 2 & Deactivate → NAT

[root@mlb-dc1-centos9 cn=config]# systemctl enable --now slapd

```
[root@mlb-dc1-centos9 cn=config]# systemctl enable --now slapd
Created symlink /etc/systemd/system/openldap.service → /usr/lib/systemd/system/slapd.service.
Created symlink /etc/systemd/system/multi-user.target.wants/slapd.service → /usr/lib/systemd/system/slapd.service.
```

2. Verify Service Status

Check that the service is running without errors.

[root@mlb-dc1-centos9 cn=config]# systemctl status slapd

```
[root@mlb-dc1-centos9 cn=config]# systemctl status slapd
● slapd.service - OpenLDAP Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/slapd.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-03-21 02:16:05 EDT; 1min 19s ago
       Docs: man:slapd
             man:slapd-config
             man:slapd-mdb
             file:///usr/share/doc/openldap-servers/guide.html
    Process: 40331 ExecStartPre=/usr/libexec/openldap/check-config.sh (code=exited, stat>
    Process: 40351 ExecStart=/usr/sbin/slapd -u ldap -h ldap:/// ldaps:/// ldapi:/// (co>
   Main PID: 40352 (slapd)
      Tasks: 2 (limit: 17150)
     Memory: 6.7M
        CPU: 159ms
     CGroup: /system.slice/slapd.service
             └─40352 /usr/sbin/slapd -u ldap -h "ldap:/// ldaps:/// ldapi:///"

Mar 21 02:16:05 mlb-dc1-centos9.csa.lk systemd[1]: Starting OpenLDAP Server Daemon...
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk runuser[40334]: pam_unix(runuser:session): sessio>
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk runuser[40334]: pam_unix(runuser:session): sessio>
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk slapd[40351]: @(#) $OpenLDAP: slapd 2.6.8 (May 19>
                                                          openldap
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk slapd[40351]: ldif_read_file: checksum error on ">
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk slapd[40351]: ldif_read_file: checksum error on ">
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk slapd[40352]: slapd starting
Mar 21 02:16:05 mlb-dc1-centos9.csa.lk systemd[1]: Started OpenLDAP Server Daemon.
```

Note: If the service fails to start, check the logs using journalctl -xeu slapd. Common issues include incorrect file permissions on /etc/openldap/slapd.d/ or the TLS certificates.

3. Verify the LDAP Service is Listening

Confirm that the server is successfully listening on the standard LDAP port (389).

[root@mlb-dc1-centos9 cn=config]# ss -nltp | grep slapd

```
[root@mlb-dc1-centos9 cn=config]# ss -nltp | grep slapd
LISTEN 0      2048         0.0.0.0:389       0.0.0.0:*    users:(("slapd",pid=40352,fd=7))
LISTEN 0      2048         0.0.0.0:636       0.0.0.0:*    users:(("slapd",pid=40352,fd=9))
LISTEN 0      2048            [::]:389          [::]:*    users:(("slapd",pid=40352,fd=8))
LISTEN 0      2048            [::]:636          [::]:*    users:(("slapd",pid=40352,fd=10))
```

9. Loading LDAP Schemas and Database Ownership

Before adding users to the directory, we must load the "Schemas" - the blueprints that define attributes like uid, gidNumber, and homeDirectory.

1. Set Database Ownership

Ensure the ldap user has full control over the database directory where the MDB files are stored.

[root@mlb-dc1-centos9 cn=config]# chown -R ldap:ldap /var/lib/ldap/

```
[root@mlb-dc1-centos9 cn=config]# chown -R ldap:ldap /var/lib/ldap/
[root@mlb-dc1-centos9 cn=config]#
```

Note: In older versions of Linux, a DB_CONFIG file was copied here. In CentOS 9, the MDB engine manages its own memory and settings automatically.

2. Add LDAP Schemas

Load the standard schemas into the running LDAP configuration using the ldapi local socket.

[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/cosine.ldif
[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/nis.ldif
[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/inetorgperson.ldif
[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/misc.ldif

```
[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/cosine.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=cosine,cn=schema,cn=config"

[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/nis.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=nis,cn=schema,cn=config"

[root@mlb-dc1-centos9 cn=config]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/inetorgperson.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=inetorgperson,cn=schema,cn=config"

[root@mlb-dc1-centos9 cn=config]#
```

```
[root@mlb-dc1-centos9 migrationtools]# ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/misc.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=misc,cn=schema,cn=config"
```

10. Configuring MigrationTools for the CSA Domain

To use the migration scripts to convert local Linux users into LDAP objects, we must first configure the common settings file to match the csa.lk domain.

1. Navigate to the MigrationTools Directory

   Enter the directory where we manually extracted the scripts in Part 2.

   [root@mlb-dc1-centos9 cn=config]# cd /usr/share/migrationtools/

```
[root@mlb-dc1-centos9 cn=config]# cd /usr/share/migrationtools/
```

2. Edit the common configuration file

   Open the migrate_common.ph file using the vi editor to define your domain settings.

   [root@mlb-dc1-centos9 migrationtools]# vi migrate_common.ph

```
[root@mlb-dc1-centos9 migrationtools]# vi migrate_common.ph
[root@mlb-dc1-centos9 migrationtools]#
```

3. Update Domain and Schema Variables

   Locate and modify the following lines (line numbers may vary slightly depending on the version, but the variable names remain the same):

- Change Default Mail Domain (Approx. Line 71):
  $DEFAULT_MAIL_DOMAIN = "csa.lk";

```
# Default DNS domain
$DEFAULT_MAIL_DOMAIN = "csa.lk";
```

- Change Default Base Name (Approx. Line 74):
  $DEFAULT_BASE = "dc=csa,dc=lk";

```
# Default base
$DEFAULT_BASE = "dc=csa,dc=lk";
```

11. Generating the Base Directory Structure (base.ldif)

Now that our configuration is complete, we must create the "Base Objects." This defines the root of our directory (dc=csa,dc=lk) and the two main folders: People (for users) and Group (for group accounts).

1. Create the base.ldif file
   Create a new file in the /root/ directory to hold these definitions.
   [root@mlb-dc1-centos9 migrationtool]# vi /root/base.ldif

```
[root@mlb-dc1-centos9 migrationtools]# vi /root/base.ldif
```

2. Add the Domain and Organizational Units
   Copy and paste the following content exactly.
   Note: Ensure there are no leading spaces at the start of any line.

dn: dc=csa,dc=lk
objectClass: top
objectClass: dcObject
objectclass: organization
o: csa lk
dc: csa

dn: cn=Manager,dc=csa,dc=lk
objectClass: organizationalRole
cn: Manager
description: Directory Manager

dn: ou=People,dc=csa,dc=lk
objectClass: organizationalUnit
ou: People

dn: ou=Group,dc=csa,dc=lk
objectClass: organizationalUnit
ou: Group

```
dn: dc=csa,dc=lk
objectClass: top
objectClass: dcObject
objectclass: organization
o: csa lk
dc: csa

dn: cn=Manager,dc=csa,dc=lk
objectClass: organizationalRole
cn: Manager
description: Directory Manager

dn: ou=People,dc=csa,dc=lk
objectClass: organizationalUnit
ou: People

dn: ou=Group,dc=csa,dc=lk
objectClass: organizationalUnit
ou: Group
```

12. Creating Local Users and Migrating to LDIF

We will now create local Linux users and groups, then use the MigrationTools we configured in Part 10 to convert them into LDAP-compatible format (LDIF).

1. Create Local Test Users

Create two test users and assign them a password.

[root@mlb-dc1-centos9 migrationtools]# useradd ldapuser1
[root@mlb-dc1-centos9 migrationtools]# useradd ldapuser2

```
[root@mlb-dc1-centos9 migrationtools]# useradd ldapuser1
[root@mlb-dc1-centos9 migrationtools]# useradd ldapuser2
```

Note: On CentOS 9, use chpasswd to set passwords via script:

[root@mlb-dc1-centos9 migrationtools]# echo "ldapuser1:redhat" | chpasswd
[root@mlb-dc1-centos9 migrationtools]# echo "ldapuser2:redhat" | chpasswd

```
[root@mlb-dc1-centos9 migrationtools]# echo "ldapuser1:redhat" | chpasswd
[root@mlb-dc1-centos9 migrationtools]# echo "ldapuser2:redhat" | chpasswd
```

2. Filter Users and Groups

In CentOS, manually created users typically start with a UID of 1000. We will extract these specific entries from the system files into temporary files in /root/.

Note: Filter users with UID 1000-1099

[root@mlb-dc1-centos9 migrationtools]# grep ":10[0-9][0-9]" /etc/passwd > /root/passwd

Note: Filter groups with GID 1000-1099

[root@mlb-dc1-centos9 migrationtools]# grep ":10[0-9][0-9]" /etc/group > /root/group

```
[root@mlb-dc1-centos9 migrationtools]# grep ":10[0-9][0-9]" /etc/passwd > /root/passwd
[root@mlb-dc1-centos9 migrationtools]# grep ":10[0-9][0-9]" /etc/group > /root/group
```

3. Convert System Files to LDIF Format

Now, use the MigrationTools Perl scripts to transform these flat text files into LDAP Data Interchange Format (LDIF).

Generate the Users LDIF:

[root@mlb-dc1-centos9 migrationtools]# PERL5LIB=. ./migrate_passwd.pl /root/passwd /root/users.ldif

Generate the Groups LDIF:

[root@mlb-dc1-centos9 migrationtools]# PERL5LIB=. ./migrate_group.pl /root/group /root/groups.ldif

```
[root@mlb-dc1-centos9 migrationtools]# PERL5LIB=. ./migrate_passwd.pl /root/passwd /root/users.ldif
[root@mlb-dc1-centos9 migrationtools]# PERL5LIB=. ./migrate_group.pl /root/group /root/groups.ldif
```

13. Importing LDIF Files into the LDAP Database

In this final step, we will push our base structure, users, and groups into the running LDAP directory.

1. Check Service Status

   Verify that the slapd service is currently running.

   [root@mlb-dc1-centos9 migrationtools]# systemctl status slapd

```
[root@mlb-dc1-centos9 migrationtools]# systemctl status slapd
● slapd.service - OpenLDAP Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/slapd.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-03-21 02:16:05 EDT; 4h 14min ago
       Docs: man:slapd
             man:slapd-config
             man:slapd-mdb
             file:///usr/share/doc/openldap-servers/guide.html
   Main PID: 40352 (slapd)
      Tasks: 4 (limit: 17150)
     Memory: 8.8M
        CPU: 326ms
     CGroup: /system.slice/slapd.service
             └─40352 /usr/sbin/slapd -u ldap -h "ldap:/// ldaps:/// ldapi:///"
```

Note: If the status is not active (running), start it using systemctl start slapd.

1. Import the Base Structure, Users, and Groups

   Run the following commands in order. Each command will prompt you for the LDAP Manager password (the one you set as olcRootPW in Part 4).

   Import the Base Domain Structure:

   [root@mlb-dc1-centos9 migrationtools]# ldapadd -x -W -D "cn=Manager,dc=csa,dc=lk" -f /root/base.ldif

```
[root@mlb-dc1-centos9 migrationtools]# ldapadd -x -W -D "cn=Manager,dc=csa,dc=lk" -f /root/base.ldif
Enter LDAP Password:
adding new entry "dc=csa,dc=lk"

adding new entry "cn=Manager,dc=csa,dc=lk"

adding new entry "ou=People,dc=csa,dc=lk"

adding new entry "ou=Group,dc=csa,dc=lk"
```

   Import the Migrated Users:

   [root@mlb-dc1-centos9 migrationtools]# ldapadd -x -W -D "cn=Manager,dc=csa,dc=lk" -f /root/users.ldif

```
[root@mlb-dc1-centos9 migrationtools]# ldapadd -x -W -D "cn=Manager,dc=csa,dc=lk" -f /root/users.ldif
Enter LDAP Password:
adding new entry "uid=svr1,ou=People,dc=csa,dc=lk"

adding new entry "uid=ldapuser1,ou=People,dc=csa,dc=lk"

adding new entry "uid=ldapuser2,ou=People,dc=csa,dc=lk"
```

Import the Migrated Groups:

[root@mlb-dc1-centos9 migrationtools]# ldapadd -x -W -D "cn=Manager,dc=csa,dc=lk" -f /root/groups.ldif

```
[root@mlb-dc1-centos9 migrationtools]# ldapadd -x -W -D "cn=Manager,dc=csa,dc=lk" -f /root/groups.ldif
Enter LDAP Password:
adding new entry "cn=svr1,ou=Group,dc=csa,dc=lk"

adding new entry "cn=ldapuser1,ou=Group,dc=csa,dc=lk"

adding new entry "cn=ldapuser2,ou=Group,dc=csa,dc=lk"
```

Note: The -x flag specifies simple authentication, and -W tells the command to prompt you for the password. Use the clear-text version of the password you encrypted earlier (e.g., redhat).

14. Testing and Verifying the LDAP Configuration

After importing the data, we must verify that the LDAP server can retrieve user information correctly. Use the ldapsearch utility to query the database.

1. Search for Specific Users

Run a targeted search to find individual user accounts.

Note: Search for ldapuser1

[root@mlb-dc1-centos9 migrationtools]# ldapsearch -x -b "dc=csa,dc=lk" "cn=ldapuser1"

```
[root@mlb-dc1-centos9 migrationtools]# ldapsearch -x -b "dc=csa,dc=lk" "cn=ldapuser1"
# extended LDIF
#
# LDAPv3
# base <dc=csa,dc=lk> with scope subtree
# filter: cn=ldapuser1
# requesting: ALL
#

# ldapuser1, People, csa.lk
dn: uid=ldapuser1,ou=People,dc=csa,dc=lk
uid: ldapuser1
cn: ldapuser1
sn: ldapuser1
mail: ldapuser1@csa.lk
objectClass: person
objectClass: organizationalPerson
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: top
objectClass: shadowAccount
userPassword:: e2NyeXB0fSQ2JE1ybmFtWE1CVXBaZXlIVk0kRzd3VHQ0Rkpiald1eG82c0FlUzF
 SYUxmTjdQN2k3V2JsMi5HMmpDWUoyZGJPb3JWVDkyejBaTjF0NFkydjFGdDNYVkxkYkM2SXN4b09v
 LzZsdUFyTjA=
shadowLastChange: 20533
shadowMax: 99999
shadowWarning: 7
loginShell: /bin/bash
uidNumber: 1001
gidNumber: 1001
homeDirectory: /home/ldapuser1
```

```
# ldapuser1, Group, csa.lk
dn: cn=ldapuser1,ou=Group,dc=csa,dc=lk
objectClass: posixGroup
objectClass: top
cn: ldapuser1
userPassword:: e2NyeXB0fXg=
gidNumber: 1001

# search result
search: 2
result: 0 Success

# numResponses: 3
# numEntries: 2
[root@mlb-dc1-centos9 migrationtools]#
```

Note: Search for ldapuser2

[root@mlb-dc1-centos9 migrationtools]# ldapsearch -x -b "dc=csa,dc=lk" "cn=ldapuser2"

```
[root@mlb-dc1-centos9 migrationtools]# ldapsearch -x -b "dc=csa,dc=lk" "cn=ldapuser2"
# extended LDIF
#
# LDAPv3
# base <dc=csa,dc=lk> with scope subtree
# filter: cn=ldapuser2
# requesting: ALL
#

# ldapuser2, People, csa.lk
dn: uid=ldapuser2,ou=People,dc=csa,dc=lk
uid: ldapuser2
cn: ldapuser2
sn: ldapuser2
mail: ldapuser2@csa.lk
objectClass: person
objectClass: organizationalPerson
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: top
objectClass: shadowAccount
userPassword:: e2NyeXB0fSQ2JHpoaGk4ZkNjVk0xNWd3eWIkWmZJZWM1ZHhhL2xEUzZRS281QlF
 XT3NBYkFQZGRBSm43bGg4NWZJdHVJMEhSbnU4N2VKY0Y4ZUpVVnl6VlMzNGdIb1VIYTAuVlVzMWFR
 TGF1Wnh0Uy8=
shadowLastChange: 20533
shadowMax: 99999
shadowWarning: 7
loginShell: /bin/bash
uidNumber: 1002
gidNumber: 1002
homeDirectory: /home/ldapuser2
```

```
# ldapuser2, Group, csa.lk
dn: cn=ldapuser2,ou=Group,dc=csa,dc=lk
objectClass: posixGroup
objectClass: top
cn: ldapuser2
userPassword:: e2NyeXB0fXg=
gidNumber: 1002

# search result
search: 2
result: 0 Success

# numResponses: 3
# numEntries: 2
[root@mlb-dc1-centos9 migrationtools]#
```

2. View the Entire Directory Tree

   To see every object currently stored in your LDAP directory (including Organizational Units, the Manager, and all Users/Groups), run a global search:

   [root@mlb-dc1-centos9 migrationtools]# ldapsearch -x -b "dc=csa,dc=lk" "(objectclass=*)"

```
[root@mlb-dc1-centos9 migrationtools]# ldapsearch -x -b "dc=csa,dc=lk" "(objectclass=*)"
```

15. Disabling the Firewall and Preparing NFS for Home Directories

To ensure clients can communicate with the LDAP server and access their personal files, we must disable the firewall and set up the Network File System (NFS).

1. Stop the Firewall

   Disable the firewall to allow LDAP (port 389) and NFS traffic without restriction.

Note: Stop the firewall service

[root@mlb-dc1-centos9 migrationtools]# systemctl stop firewalld

2. Preparing for Home Directory Sharing

   LDAP only handles the authentication (username and password). It does not automatically move a user's files to a client machine. To ensure users can access their files regardless of which server they log into, we must share the /home directory via NFS.

Important Note: If a user logs into a client machine and their home directory is not available, they will receive the error: "Home directory not found, logging in with HOME=/".

3. Why use NFS?

- Centralized Storage: Users can save their data remotely on the LDAP Server.
- Consistency: Their desktop settings, documents, and scripts follow them to any client machine in the csa.lk domain.

16. NFS Configuration to Export the Home Directory

To ensure LDAP users can access their files from any client machine, we must export the /home directory using the Network File System (NFS).

1. Configure the NFS Exports File

   Define which directories will be shared across the network.

   [root@mlb-dc1-centos9 ~]# vi /etc/exports

```
[root@mlb-dc1-centos9 ~]# vi /etc/exports
[root@mlb-dc1-centos9 ~]#
```

   Add the following line to the file:

   /home *(rw,sync)

```
/home *(rw,sync)
~
~
```

Note: rw allows read/write access, sync ensures data is written to disk before confirming.

2. Install and Start NFS Services

   Ensure the necessary NFS utilities are installed and the services are active.

Note: Install NFS utilities

Activate → NAT & Deactivate → VMnet2

[root@mlb-dc1-centos9 ~]# dnf -y install rpcbind nfs-utils

```
[root@mlb-dc1-centos9 ~]# dnf -y install rpcbind nfs-utils
Last metadata expiration check: 3:19:07 ago on Sat 21 Mar 2026 04:00:46 AM EDT.
Dependencies resolved.
================================================================================================================
 Package                          Architecture        Version                     Repository          Size
================================================================================================================
Installing:
 nfs-utils                        x86_64              1:2.5.4-42.el9              baseos             462 k
 rpcbind                          x86_64              1.2.6-7.el9                 baseos              58 k
Upgrading:
 libipa_hbac                      x86_64              2.9.8-1.el9                 baseos              33 k
 libldb                           x86_64              4.23.5-6.el9                baseos             186 k
 libsmbclient                     x86_64              4.23.5-6.el9                baseos              75 k
 libsss_certmap                   x86_64              2.9.8-1.el9                 baseos              87 k
 libsss_idmap                     x86_64              2.9.8-1.el9                 baseos              38 k
 libsss_nss_idmap                 x86_64              2.9.8-1.el9                 baseos              43 k
 libsss_sudo                      x86_64              2.9.8-1.el9                 baseos              32 k
 libtalloc                        x86_64              2.4.3-1.el9                 baseos              34 k
 libtdb                           x86_64              1.4.14-1.el9                baseos              53 k
 libtevent                        x86_64              0.17.1-1.el9                baseos              50 k
 libwbclient                      x86_64              4.23.5-6.el9                baseos              43 k
```

Note: Start the services:

[root@mlb-dc1-centos9 ~]# systemctl start rpcbind

[root@mlb-dc1-centos9 ~]# systemctl start nfs-server

```
[root@mlb-dc1-centos9 ~]# systemctl start rpcbind
[root@mlb-dc1-centos9 ~]# systemctl start nfs-server
```

Note: Enable and start the services

[root@mlb-dc1-centos9 ~]# systemctl enable rpcbind

[root@mlb-dc1-centos9 ~]# systemctl enable nfs-server

```
[root@mlb-dc1-centos9 ~]# systemctl enable rpcbind
[root@mlb-dc1-centos9 ~]# systemctl enable nfs-server
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/system/nfs-server.service.
[root@mlb-dc1-centos9 ~]#
```

3. Test the NFS Configuration

   Verify that the /home directory is correctly exported and visible to the network.

   Activate → VMnet2 & Deactivate → NAT

   [root@mlb-dc1-centos9 ~]# showmount -e

```
[root@mlb-dc1-centos9 ~]# showmount -e
Export list for mlb-dc1-centos9.csa.lk:
/home *
```

Client End Configuration (Fedora 39)

Follow these steps to connect your Fedora 39 client to the LDAP server and enable user home directories.

1. Install Required Packages

   Fedora 39 uses sssd to handle LDAP communication.

   [root@ndm-cli-fedora39 ~]# dnf install -y sssd sssd-ldap oddjob-mkhomedir openldap-clients

```
root@ndm-cli-fedora39:/home/client1# dnf install -y sssd sssd-ldap oddjob-mkhomedir openldap-clients
Last metadata expiration check: 0:32:25 ago on Sat 21 Mar 2026 04:58:48 PM +0530.
Package sssd-2.9.2-1.fc39.x86_64 is already installed.
Package sssd-ldap-2.9.2-1.fc39.x86_64 is already installed.
Package oddjob-mkhomedir-0.34.7-9.fc39.x86_64 is already installed.
Package openldap-clients-2.6.7-1.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@ndm-cli-fedora39:/home/client1#
```

2. Configure Authentication Profile

   Instead of the old blue "TUI" screen, Fedora uses authselect. This command tells the system to use SSSD for logins and automatically create home directories for new users.

   [root@ndm-cli-fedora39 ~]# authselect select sssd with-mkhomedir --force

```
root@ndm-cli-fedora39:/home/client1# authselect select sssd with-mkhomedir --force
Backup stored at /var/lib/authselect/backups/2026-03-21-12-01-55.zzEz70
Profile "sssd" was selected.

Make sure that SSSD service is configured and enabled. See SSSD documentation for more information.

- with-mkhomedir is selected, make sure pam_oddjob_mkhomedir module
  is present and oddjobd service is enabled and active
  - systemctl enable --now oddjobd.service
```

   [root@ndm-cli-fedora39 ~]# systemctl enable --now oddjobd.service

```
root@fedora:/home/client1# systemctl enable --now oddjobd.service
Created symlink /etc/systemd/system/multi-user.target.wants/oddjobd.service → /usr/lib/systemd/system/oddjobd.service.
```

3. Configure SSSD to talk to your LDAP Server

   You need to manually create the configuration file for the LDAP connection.

   1. Create the file:
      [root@ndm-cli-fedora39 ~]# vi /etc/sssd/sssd.conf

      ```
      root@ndm-cli-fedora39:/home/client1# vi /etc/sssd/sssd.conf
      root@ndm-cli-fedora39:/home/client1#
      ```

   2. Paste the following configuration (Ensure the ldap_uri and ldap_search_base match your server details):

      [sssd]
      services = nss, pam
      domains = default

      [domain/default]
      id_provider = ldap
      ldap_uri = ldap://10.0.1.2
      ldap_search_base = dc=csa,dc=lk
      auth_provider = ldap
      chpass_provider = ldap
      ldap_tls_reqcert = allow
      cache_credentials = True

      ```
      [sssd]
      services = nss, pam
      domains = default

      [domain/default]
      id_provider = ldap
      ldap_uri = ldap://10.0.1.2
      ldap_search_base = dc=csa,dc=lk
      auth_provider = ldap
      chpass_provider = ldap
      ldap_tls_reqcert = allow
      cache_credentials = True
      ```

   3. Set Permissions & Restart: SSSD will refuse to start if the file permissions are not strictly set to 600.
      [root@ndm-cli-fedora39 ~]# chmod 600 /etc/sssd/sssd.conf
      [root@ndm-cli-fedora39 ~]# systemctl restart sssd

      ```
      root@ndm-cli-fedora39:/home/client1# chmod 600 /etc/sssd/sssd.conf
      root@ndm-cli-fedora39:/home/client1# systemctl restart sssd
      ```

4. Test the Client Configuration

   Check if the client can see the LDAP user:
   [root@ndm-cli-fedora39 ~]# getent passwd ldapuser1
   [root@ndm-cli-fedora39 ~]# getent passwd ldapuser2

```
root@ndm-cli-fedora39:/home/client1# getent passwd ldapuser1
ldapuser1:*:1001:1001:ldapuser1:/home/ldapuser1:/bin/bash
root@ndm-cli-fedora39:/home/client1# getent passwd ldapuser2
ldapuser2:*:1002:1002:ldapuser2:/home/ldapuser2:/bin/bash
```

5. Mount the LDAP Users' Home Directory

   To ensure users can access their files, add the network mount to your filesystem table.

   1. Edit the fstab file:

      [root@ndm-cli-fedora39 ~]# vi /etc/fstab

      ```
      root@ndm-cli-fedora39:/home/client1# vi /etc/fstab
      ```

   2. Add this line at the very bottom:

      10.0.1.2:/home /home nfs defaults 0 0

      ```
      10.0.1.2:/home /home nfs defaults 0 0
      ~
      ```

   3. Mount the directory:

      [root@ndm-cli-fedora39 ~]# mount -a

      ```
      root@ndm-cli-fedora39:/home/client1# mount -a
      ```

   4. Verify the Mount

      Run this command to confirm the Fedora client is successfully seeing the CentOS server's storage.

      [root@ndm-cli-fedora39 ~]# df -h | grep /home

      ```
      root@ndm-cli-fedora39:/home/client1# df -h | grep /home
      10.0.1.2:/home    17G  6.6G   11G  39% /home
      ```

6. Final Login Test

   Try switching to the LDAP user to ensure everything (authentication + home directory) is working:

   [root@ndm-cli-fedora39 ~]# su - ldapuser1
   [ldapuser1@ndm-cli-fedora39 ~]# exit

   ```
   root@ndm-cli-fedora39:/home/client1# su - ldapuser1
   Last login: Sun Mar 22 02:15:57 +0530 2026 on pts/1
   [ldapuser1@ndm-cli-fedora39 ~]$ exit
   logout
   ```

   [root@ndm-cli-fedora39 ~]# su - ldapuser2
   [ldapuser2@ndm-cli-fedora39 ~]# exit

   ```
   root@ndm-cli-fedora39:/home/client1# su - ldapuser2
   Last login: Sun Mar 22 02:16:04 +0530 2026 on pts/1
   [ldapuser2@ndm-cli-fedora39 ~]$ exit
   logout
   ```

7. Unmounting the Network Share

   Note: If you need to return to your local Fedora home directories, use a "Lazy Unmount":

   [root@ndm-cli-fedora39 ~]# umount -l /home

```
root@ndm-cli-fedora39:/home/client1# umount -l /home
root@ndm-cli-fedora39:/home/client1#
```

Note: The -l flag ensures the unmount happens even if the system thinks the directory is "busy."

Note to Students: Try the following steps by yourself.

ADDITIONAL: GUI Administration with phpLDAPadmin

Objective: Since managing LDAP via command line can be complex, we will use a web-based GUI tool called phpLDAPadmin.

8. Install phpLDAPadmin

   On the CentOS Server, install the EPEL repository and the required packages

   [root@mlb-dc1-centos9 ~]# dnf install epel-release -y

   [root@mlb-dc1-centos9 ~]# dnf install httpd phpldapadmin -y

9. Configure Web Access Permissions

   By default, phpLDAPadmin only allows access from the server itself. We must allow our Fedora client to connect.

   1. Open the Apache config for phpLDAPadmin

      [root@mlb-dc1-centos9 ~]# vi /etc/httpd/conf.d/phpldapadmin.conf
   2. Look for the **<Directory /usr/share/phpldapadmin/htdocs>** section.
   3. Change **Require local** to **Require all granted**. (Approx. Line 11)
   4. Save and exit.

10. Update phpLDAPadmin Configuration

    Match the web tool settings to your specific LDAP domain and IP.

    1. Open the config file

       [root@mlb-dc1-centos9 ~]# vi /etc/phpldapadmin/config.php
    2. Search for the "Define LDAP Servers" section and update these values:
       - Server Name: **$servers->setValue('server','name','CSA LDAP Server');** (Approx. Line 322)
       - Server Host: **$servers->setValue('server','host','192.168.76.129');** (Approx. Line 329)

         Note: The IP address 192.168.76.129 used in this step is the Server's NAT IP. If your VM has assigned a different IP, you must replace 192.168.76.129 with the result of your ip addr command or ifconfig command.

       - Base DN: **$servers->setValue('server','base',array('dc=csa,dc=lk'));** (Approx. Line 336)
       - Bind ID: **$servers->setValue('login','bind_id','cn=Manager,dc=csa,dc=lk');** (Approx. Line 454)

11. Start Services and Configure Firewall

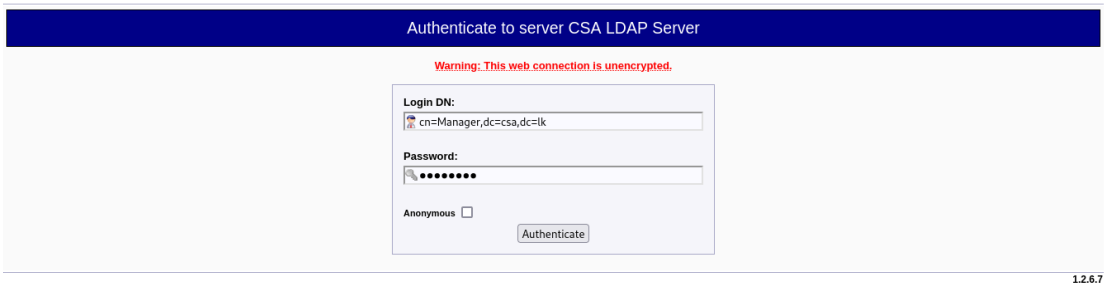
    Enable the web server and ensure the firewall allows traffic.

[root@mlb-dc1-centos9 ~]# systemctl enable --now httpd
[root@mlb-dc1-centos9 ~]# firewall-cmd --permanent --add-service=http
[root@mlb-dc1-centos9 ~]# firewall-cmd --reload
[root@mlb-dc1-centos9 ~]# setsebool -P httpd_can_connect_ldap on

12. Access the GUI via Web Browser
    1. Open the Firefox web browser on your Fedora Client.
    2. In the address bar, enter the following URL (Replace with your Server's actual IP if it is different)
       http://192.168.76.129/phpldapadmin
    3. The phpLDAPadmin welcome screen should appear.
       - Verification: On the left-hand menu, you should see your server name: "**CSA LDAP Server**".

       - Login: Click the "**login**" link under the server name.

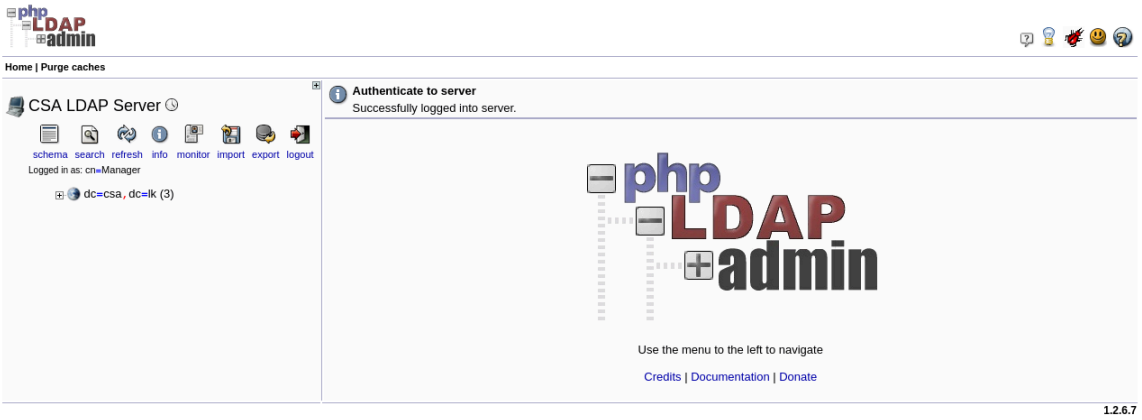
       - Credentials: * Login DN: **cn=Manager,dc=csa,dc=lk**
         - Password: (Enter the LDAP password you created in Step 3)



13. Exploring the phpLDAPadmin Main Console

Once you have successfully logged in, the main console screen of phpLDAPadmin will open.

1. Verify the Domain Connection: In the left-hand navigation pane, you will see your server listed as "**CSA LDAP Server**".
2. Expand the Tree: Click the "**+**" icon next to the server name and the domain **dc=csa,dc=lk**.



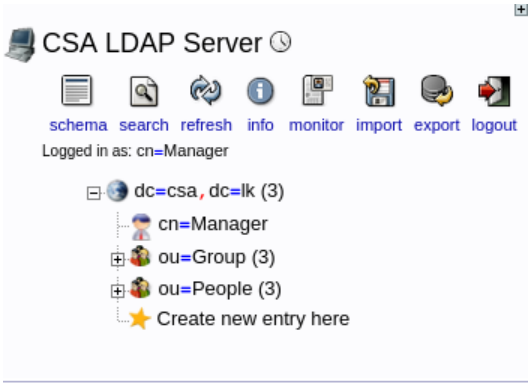
3. Management Capabilities: This console is your visual gateway to the LDAP database. From here, you can easily manage the directory structure without using complex command-line syntax:
   - Organizational Units (OU): Create departments like Sales, HR, or IT.
   - Users: Add new employees or system users with specific attributes.
   - Groups: Create security groups to manage permissions across the network.

Verification Task: Ensure you can see the existing LDAP users (like ldapuser1 and ldapuser2) that you migrated earlier in the lab. They should appear under the ou=People or dc=csa,dc=lk branch.

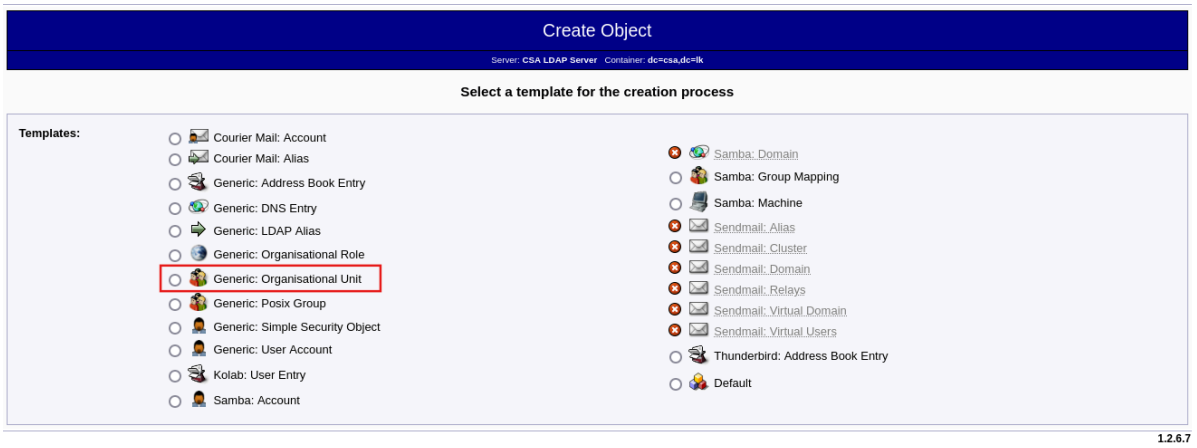
Sample Configuration: Testing the GUI Interface

Objective: Let's create some sample objects using the phpLDAPadmin interface and verify that they are correctly added to the LDAP server database.

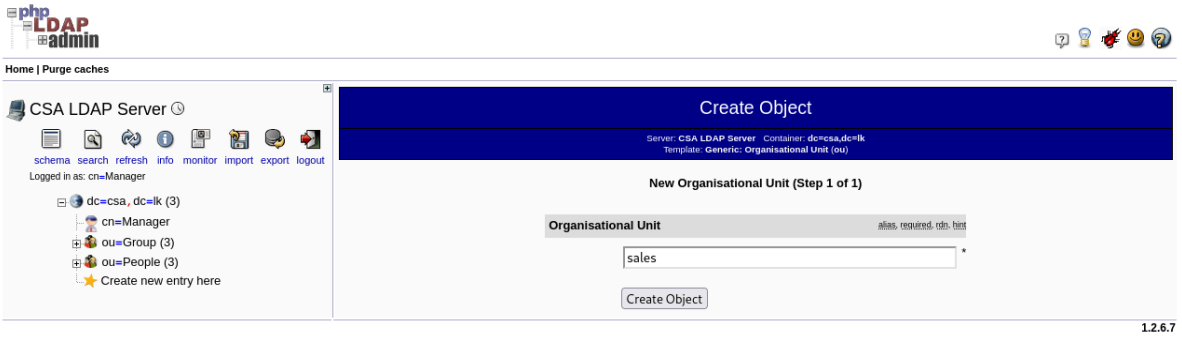
1. Start the Process: In the left-hand navigation pane, click on the "+" sign next to your domain: dc=csa,dc=lk.
2. Initiate New Entry: Click the link labeled "**Create new entry here**" located directly under your domain name.



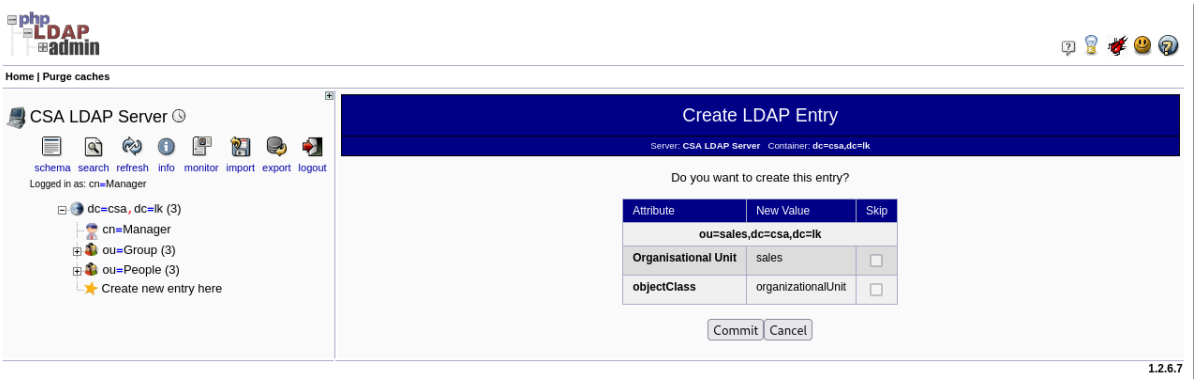
3. Create an Organizational Unit (OU):
   - From the list of templates, select "**Generic: Organizational Unit**".



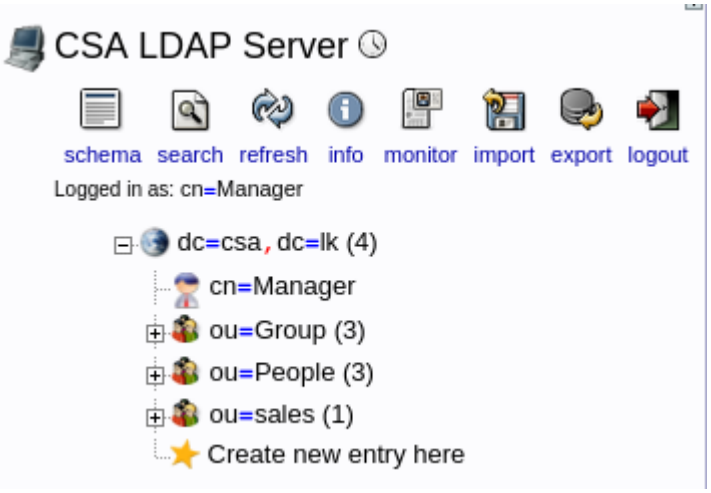
   - **OU Name**: Enter **sales**.



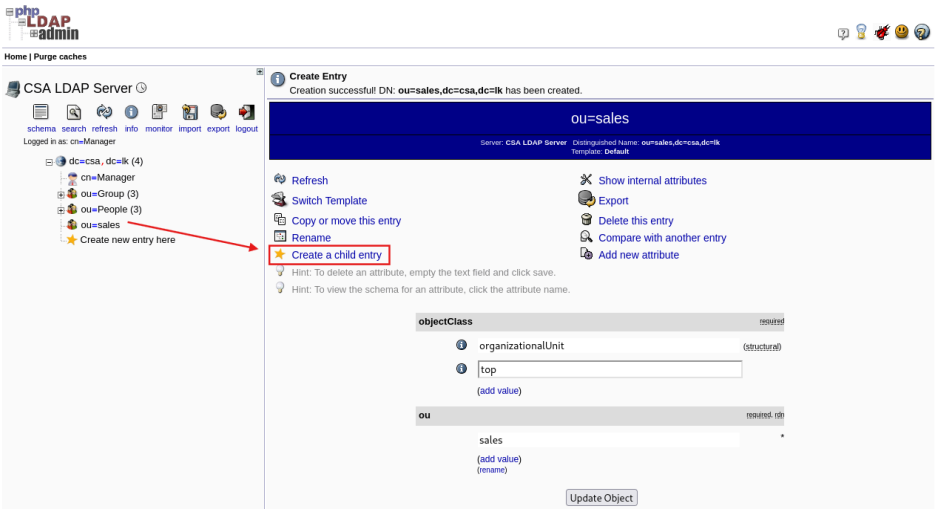
   - Click the "**Create Object**" button, then click "**Commit**" to save it to the database.



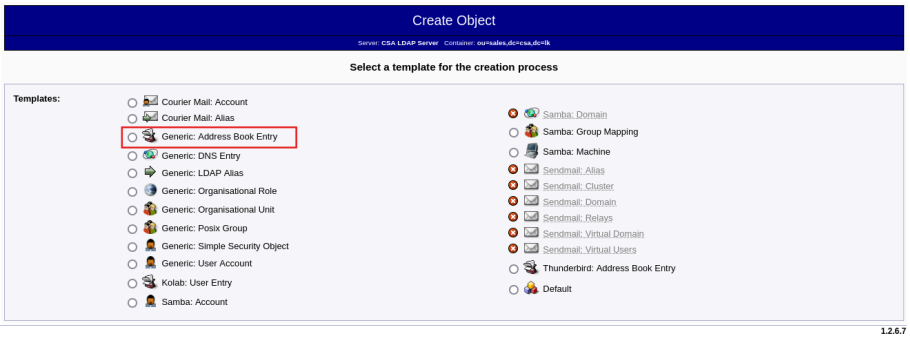
- Verification: You should now see ou=sales appear in the tree on the left.



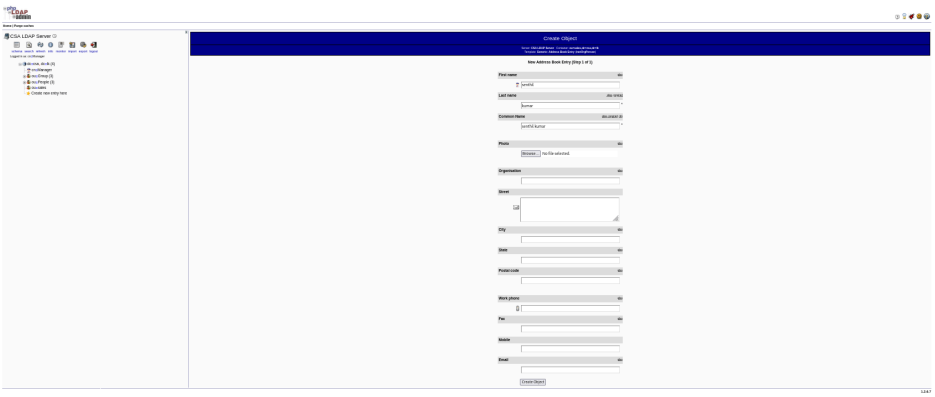
4. Create a User (Child Entry):
   - Click on your newly created **ou=sales** folder.
   - Click "**Create a child entry**".



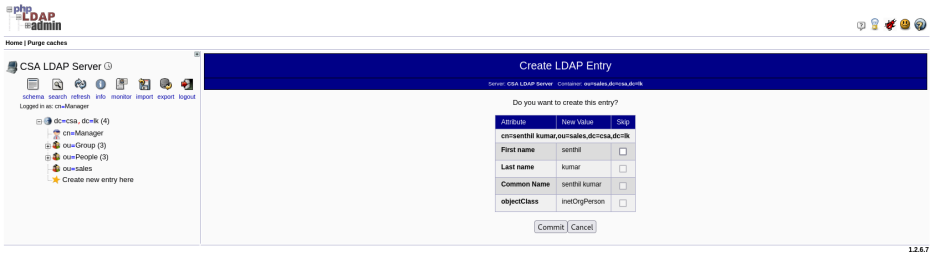
- Select "**Generic: Address book entry**".



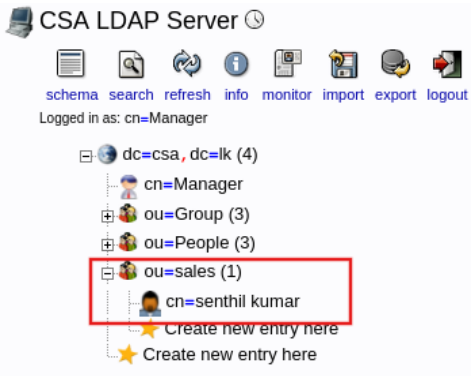
- **First name: senthil**
- **Last name: kumar**
- **Common Name (cn): senthil kumar**



- Click "**Create Object**", then "**Commit**".



- The newly created user "**senthil kumar**" is now officially a child object of the "**sales**" Organizational Unit (OU).



Final Verification Challenge

1. Verify GUI changes via the CentOS Command Line To prove that the web interface is actually writing to your LDAP database, go back to your CentOS Server terminal and run a search.

[root@mlb-dc1-centos9 ~]# ldapsearch -x -b "dc=csa,dc=lk" "(cn=senthil*)"

Note: Expected Result: The terminal should output the full LDAP record for Senthil Kumar, showing his dn (Distinguished Name) as: **dn: cn=senthil kumar,ou=sales,dc=csa,dc=lk**